

Placing /aw/ Retraction in the Retreat from the Southern Vowel Shift in Raleigh, North Carolina

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journals.sagepub.com/home/eng**Marie Bissell** 

Abstract

This study focuses on changes over apparent time in the /aw/ vowel class (e.g., MOUTH) in Raleigh, a city in North Carolina whose speakers have undergone vocalic changes away from the Southern Vowel Shift in response to extensive in-migration from the northern United States. Although Southerners have conventionally fronted /aw/ nuclei along with other speakers in the Southeastern super-region, some researchers have noted a trend of /aw/ nucleus retraction among younger speakers in urban areas, such as Philadelphia and Kansas City. I examine how the acoustics of /aw/ production have shifted generationally in Raleigh, with a particular focus on nucleus retraction and glide height. The analysis suggests that concurrent phonologically-driven and socially-driven forces may be acting upon this vowel class in the speech of white Raleigh residents.

Keywords

vowel dynamics, dialect contact, American Englishes

1. Introduction

Raleigh, North Carolina, has been the subject of several sociophonetic studies focusing on changes involving the Southern Vowel Shift (SVS), primarily due to its history of extensive in-migration from the northern United States. Since the founding of the technology hub Research Triangle Park in 1959, more and more non-native North

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Carolínians have relocated to the Raleigh-Durham metropolitan area to take jobs at companies like IBM, Microsoft, and Cisco (Rohe 2011). Dodsworth and Kohn (2012) report sizeable generational differences in how lifelong white residents of Raleigh produced front vowels impacted by the SVS, especially in the cases of /æ/ (e.g., TRAP) in Stage 1 and the mid front vowels /e ε/ (e.g., FACE, DRESS) in Stage 2. They find minimal evidence of high front vowel reversal (i.e., Stage 3) in their corpus, in which /i ɪ/ (e.g., FLEECE, KIT) nearly switch acoustic targets. Taken together with Labov, Ash, and Boberg's (2006) finding of only one Raleigh speaker with high front vowel reversal, Dodsworth and Kohn's (2012) results suggest that all of the stages of the SVS may not have diffused to the Raleigh area. The city's dialectological setting is one suited to observing vocalic changes in progress, especially those implicated in what was likely an incomplete uptake of the SVS.

Changes in back vowels in the area have been underexplored due to their involvement in large-scale changes in North American English varieties not unique to the South (cf. Dodsworth & Kohn 2012). Even less attention has been paid to the /aw/ vowel class (e.g., MOUTH), which has lower lexical frequency than both /o/ and /u/ (e.g., GOAT, GOOSE) in English (Nusbaum, Pisoni & Davis 1984). The current study focuses on providing a detailed analysis of /aw/ in Raleigh to illuminate its role in and relationship to other ongoing changes in the area involving local versus supra-regional vocalic norms.

In the current study, I investigate how apparent time changes in /aw/ productions among Raleigh, North Carolina, speakers reflect tensions between local and supra-regional vocalic patterns. I consider both social and phonological factors that may prompt the development of such changes to better conceptualize their motivations, especially in the specific context of existing research on speakers in other cities in the Southeastern super-region (Labov, Rosenfelder & Fruehwald 2013; Strelluf 2018).

First, in section 2, I summarize the existing literature about the /aw/ vowel class and its role in defining Labov, Ash, and Boberg (2006)'s Southeastern super-region. In particular, I focus on recent reports of /aw/ nucleus retraction (i.e., lowering and backing) in Kansas City (Strelluf 2018) and Philadelphia (Labov, Rosenfelder & Fruehwald 2013). In section 3, I detail the methods for acoustic analysis of the speech data corpus from Raleigh. Then, in section 4, I analyze acoustic data from lifelong white residents of Raleigh to investigate whether and how /aw/ nucleus retraction has been implemented over apparent time. Finally, section 5 discusses the potential relationships between Raleigh speakers' retreat from the Southern Vowel Shift and participation in the novel trend of /aw/ nucleus retraction in the Southeastern super-region.

2. Background

While researchers have agreed that processes involving front vowels, along with /aɪ/ monophthongization (e.g., PRICE), should be included under the broad SVS umbrella, how to treat back vowels has generated some controversy. Labov, Yaeger, and Steiner (1972) included back vowel fronting (i.e., lowered F2 values for /o u/) in their definition of the SVS, but later Labov, Ash, and Boberg (2006) excluded back vowel

fronting from their definition of the SVS due to its widespread presence in American Englishes (cf. Dodsworth & Kohn 2012). The process of /aw/ fronting has been noted to occur in tandem with other SVS features for both white speakers (Baranowski 2008; Pederson 1986; Thomas 2003) and African American speakers (Thomas, 2001). Additionally, Labov, Ash, and Boberg (2006) described /aw/ fronting as being characteristic of the larger Southeastern super-region, encompassing the Midland, the South, and the Mid-Atlantic dialect regions.

Previous studies of other cities in the Southeastern super-region indicate that /aw/ is beginning to retract for younger speakers over apparent time in certain urban areas, starting with speakers born in the 1950s (e.g., Labov 1966; Bailey, Wikle, Tillery & Sand 1991). Labov, Rosenfelder, and Fruehwald (2013) observe /aw/ fronting among older white speakers, but also a trend toward reversal among younger white speakers in the Philadelphia Neighborhood Corpus. They attribute /aw/ retraction to social factors, theorizing that younger Philadelphians are trying to avoid sounding Southern. Strelluf (2018) also finds a reversal of /aw/ fronting among younger white speakers in Kansas City, Missouri. He proposes that younger Kansas Citians may have reassigned the nucleus of /aw/ from /a/ to /æ/, such that /aw/ then begins to pattern with /æ/ (i.e., retracting) as opposed to with /u/ and /o/ (i.e., fronting). Taken together, these findings suggest that white speakers in multiple areas within the Southeastern super-region are beginning to reverse /aw/ fronting, which has been described as a keystone feature of this dialect area (Labov, Ash & Boberg 2006). As a Southern city (as opposed to Philadelphia, a Mid-Atlantic city, and Kansas City, a Midland city), Raleigh has the potential to reveal whether the reversal of /aw/ fronting is part of a more widespread supra-regional phenomenon among white speakers. Additionally, Pederson (1986) and Thomas (2003) note that /aw/ productions in white Southern English tend to have lowered glides, which could also be implicated in orientation toward more supra-regional norms.

Aside from figuring out whether their motivations are social or phonological in nature, what is worth noting about these apparent time changes in /aw/ is the implications they may have for conceptualizing the relationships between local vocalic systems and larger regional patterns. While Labov, Rosenfelder, and Fruehwald (2013) argue that reversal of /aw/ fronting among white speakers in Philadelphia signifies movement away from something stigmatized (i.e., sounding Southern), Strelluf's (2018) interpretation of this reversal among white speakers in Kansas City includes movement both away from Midland norms and, critically, toward norms of other dialect regions, like the West. These discussions mirror those happening elsewhere with respect to other vocalic changes in American English varieties, such as Becker's (2019) work on the low back merger shift as movement away from local norms versus orientation toward a supra-regional system.

3. Methods

The data analyzed in this study comes from the Raleigh Corpus (Dodsworth & Kohn 2012), a large collection of semi-structured conversational interviews in English

with lifelong white residents of Raleigh, North Carolina. Participants were interviewed either in a university office or in their own homes, and the topics covered in the interviews ranged widely but generally included discussions of what it was like to grow up in the Raleigh area and changes the participants had experienced over their lifetimes. The audio data from these interviews was forced-aligned using P2FA (Yuan & Liberman 2008) and then hand-corrected where necessary by acoustically-trained researchers (Dodsworth & Kohn 2012). In particular, visually obvious errors in segmentations were fixed. Although data collection for the corpus is still ongoing, in the current study I examine data from 104 speakers. Ten speakers from the overall data set of 114 speakers were excluded due to belonging to the unskilled white collar class, which contained too few speakers for meaningful statistical analysis. The remaining speakers were classified as either blue collar or white collar, with blue collar speakers reporting an occupation involving manual labor and white collar speakers reporting an occupation without (Dodsworth & Kohn 2012). Birth years for the speakers included in this analysis range from 1923 to 1996 ($M=1963$). Table 1 shows other relevant demographic information for the speakers included in this analysis.

Table 1. Gender and Class Distribution of Speakers in this Analysis

Social characteristics	Women	Men	Totals
Blue collar	5	10	15
White collar	44	45	89
Totals	49	55	104

I analyze /aw/ tokens ($N=6571$) preceding obstruents, a class that includes plosives (e.g., /t/ in *shout*), fricatives (e.g., /s/ in *house*), and affricates (e.g., /tʃ/ in *couch*). Tokens in other environments were excluded due to high levels of variability in pre-nasal and pre-lateral vocalic productions elsewhere in the same corpus (Dodsworth & Kohn 2012). F1 (Hz) estimates at 25 percent (nucleus) and 75 percent (glide), and F2 (Hz) estimates at 25 percent (nucleus) segment durations were automatically extracted using a script in Praat (Boersma & Weenink 2021).

I fit a linear mixed-effects regression model predicting each of the three dependent variables (F1 NUCLEUS, F2 NUCLEUS, and F1 GLIDE) using BIRTH YEAR, GENDER (woman, man), CLASS (blue collar, white collar), FOLLOWING VOICE (voiceless, voiced), and their interactions in a fully factorial design. The categorical predictors are dummy-coded, such that the first level alphabetically gets assigned as the reference level for the purpose of fitting the models. DURATION is also included as a covariate in each model. All numerical predictors and covariates are scaled to be centered on zero to facilitate fitting the regression models. I fit the models using the lme4 package (Bates, Mächler, Bolker & Walker 2015) in R software (R Core Team 2021).

BIRTH YEAR is included as a predictor due to evidence that Raleigh speakers are retreating from the hallmark vocalic patterns of the SVS for front vowels (Dodsworth & Kohn 2012; Dodsworth, 2013; Dodsworth & Benton 2017). GENDER is included as a predictor because women have been shown to lead ongoing vocalic changes from above, at least in western contexts (Frazer 1983; Moonwomon 1989; Labov 1990; Fought 1999; Schilling-Estes 2002), although Dodsworth and Kohn (2012) reported few gender effects for changes in front vowels among speakers in this same corpus. The CLASS predictor accounts for its reported role in changes in progress (Labov 1966, 1990), especially in the context of the city of Raleigh (Dodsworth 2009, 2013). The intertwined topics of class or occupation and social network structures have been carefully explored elsewhere in work on Raleigh (Dodsworth 2014, 2019; Dodsworth & Benton 2017), although social network structures are beyond the scope of the current study. The FOLLOWING VOICE predictor accounts for known impacts on the acoustics of /aw/ productions (Peterson & Lehiste 1960; Gay 1968; Moreton 2004; Pycha & Dahan 2016; Onosson 2018). This predictor is also included to capture any pre-voiceless /aw/ raising, if present, especially considering recent evidence suggesting that pre-voiceless /aɪ/ raising occurs among some younger speakers in Raleigh (Dodsworth, Forrest & Kohn 2020). The pre-voiceless /aɪ/ raising that has been reported in the United States has been termed “American Raising” (Davis & Berkson 2021), distinct from the concomitant pre-voiceless /aɪ/ and /aw/ raising that is widespread in Canada (thus, termed “Canadian Raising”).

The maximal design-driven random effects structure supported by the complexity of the data is used to build each model (Barr, Levy, Scheepers & Tily 2013). For all three models shown in the results section, this random effects structure consists of random by-speaker and by-word intercepts. This final random effects structure is determined by first attempting to fit all potential predictor slopes over by-word and by-subject random intercepts, and then removing the predictor slopes with the smallest variance one-by-one until the model converges without errors. The significance of each predictor in the model is evaluated with log-likelihood comparisons ($\alpha = .05$).

4. Results

4.1. F2 Nucleus Model

The log-likelihood comparisons for the F2 nucleus model, shown in Table 2, reveal a significant main effect of BIRTH YEAR, such that younger speakers have lower F2 nucleus values (i.e., more backing) than older speakers. Figure 1 demonstrates this linear relationship between BIRTH YEAR and F2 NUCLEUS values, in which the youngest speakers have the lowest F2 nucleus values. The DURATION covariate is also significant, such that longer productions have higher F2 nucleus values (i.e., less backing) than shorter productions.

Table 2. Log-likelihood Comparisons for the Linear Mixed-effects Model with F2 Nucleus as Dependent Variable

Coefficient	χ^2 Value	Degrees of freedom	p Value
BIRTH YEAR $\times$ GENDER $\times$ CLASS	0.72	1	n.s.
BIRTH YEAR $\times$ GENDER	0.15	1	n.s.
BIRTH YEAR $\times$ CLASS	0.02	1	n.s.
GENDER $\times$ CLASS	1.06	1	n.s.
BIRTH YEAR	68.36	1	<.001
GENDER	3.12	1	n.s.
CLASS	0.11	1	n.s.
FOLLOWING VOICE	2.51	1	n.s.
DURATION	212.27	1	<.001

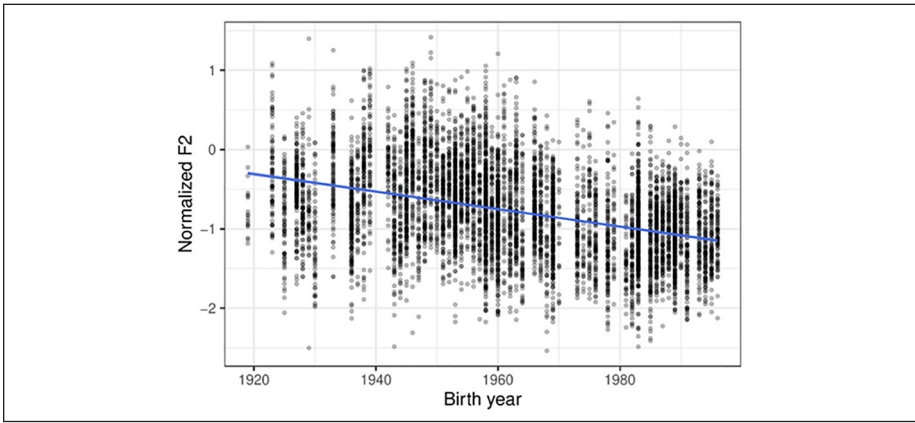


Figure 1. Linear Effect of BIRTH YEAR on F2 Nucleus Values

The process of /aw/ retraction appears to be well underway in the Raleigh area. The results of this model support my prediction that younger speakers are backing /aw/ nuclei, matching behaviors of younger speakers in the Southeastern super-region in Philadelphia (Labov, Rosenfelder & Fruehwald 2013) and Kansas City (Strelluf 2018). While a linear model does not permit fitting non-linear predictors, a function that models non-linear effects of BIRTH YEAR on F2 nucleus values is instructive for discerning the finer-grained details of how this change unfolds. Figure 2 shows a loess curve modeling the BIRTH YEAR-F2 NUCLEUS relationship, suggesting that /aw/ nuclei are fronting up until around speakers born in 1950 or so before beginning to back. The cross-over-point of the two loess curves in Figure 2, at which point women become more advanced in /aw/ retraction than same-aged men, occurs in the middle of the generation of the first Raleigh speakers with near-lifelong exposure to Northern dialects. The timing of this crossover matches the timing of other SVS-related vocalic changes reported by Dodsworth and Kohn (2012), indicating that these changes may be part of a broader dialectological shift in Raleigh speakers' vocalic systems. The general trend

in this data also closely mirrors Labov, Rosenfelder, and Fruehwald’s (2013) findings among Philadelphians. There, older speakers front /aw/ nuclei prior to younger speakers eventually backing them. In particular, speakers born in the 1950s are at the forefront of this reversal of fronting in both the Raleigh data and the Philadelphia data, pointing to potential parallelism in phonological or social constraints across the Southeastern super-region.

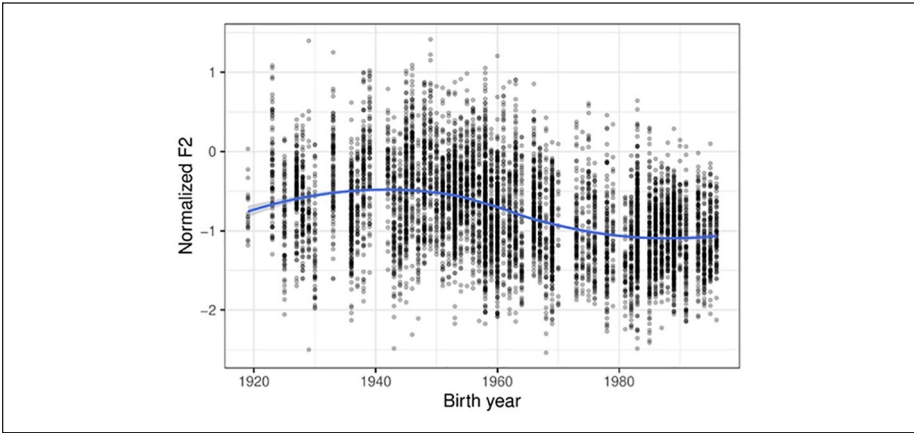


Figure 2. Loess Curve of the Effect of BIRTH YEAR on F2 Nucleus Values

4.2. F1 Nucleus Model

The log-likelihood comparisons for the F1 nucleus model, shown in Table 3, demonstrate a significant main effect of BIRTH YEAR: younger speakers have higher F1 nucleus values (i.e., more lowering) than older speakers. Figure 3 shows the linear relationship between BIRTH YEAR and F1 nucleus values, in which the youngest speakers have the highest F1 nucleus values. That is, there is a trend of nucleus lowering over apparent time.

Table 3. Log-likelihood Comparisons for the Linear Mixed-Effects Model With F1 Nucleus as Dependent Variable

Coefficient	χ^2 Value	Degrees of freedom	<i>p</i> Value
BIRTH YEAR $\times$ GENDER $\times$ CLASS	0.02	1	n.s.
BIRTH YEAR $\times$ GENDER	0.83	1	n.s.
BIRTH YEAR $\times$ CLASS	0.40	1	n.s.
GENDER $\times$ CLASS	0.31	1	n.s.
BIRTH YEAR	3.98	1	.046
GENDER	10.42	1	.001
CLASS	7.86	1	.005
FOLLOWING VOICE	0.34	1	n.s.
DURATION	486.07	1	<.001

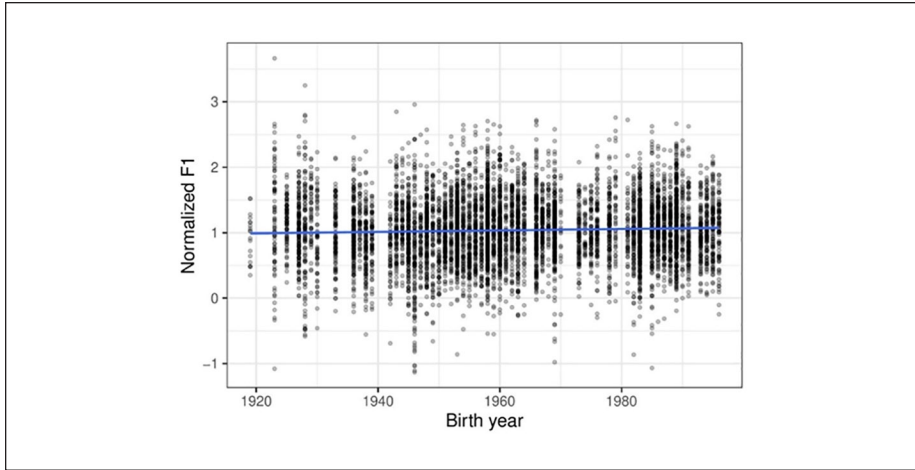


Figure 3. Linear Effect of BIRTH YEAR on F1 Nucleus Values

The results shown in Table 3 also indicate a significant main effect of *GENDER*, such that men have higher F1 nucleus values (i.e., lower nuclei) than women. Since the overall trend in the model is nucleus lowering over apparent time, this result suggests that men are consistently more advanced in this change than same-aged women. There is also a significant main effect of *CLASS*, such that blue collar speakers have higher F1 nucleus values (i.e., lower nuclei) than white collar speakers. This result suggests that blue collar speakers are leading nucleus lowering in this community relative to white collar speakers, as well.

Finally, the model shows a significant *DURATION* covariate: longer vowel productions have lower nuclei than shorter vowel productions. However, there is no significant effect of *FOLLOWING VOICE*, suggesting that the pre-voiceless /aɪ/ raising reported by Dodsworth, Forrest, and Kohn (2020) is not also occurring for the /aw/ diphthong. Although pre-voiceless /aw/ raising has rarely been reported in the United States (Carmichael 2020; Bissell & Carmichael 2022), it is worth noting that changes in /aw/ nucleus height among white Raleigh speakers do not appear to be phonologically conditioned by following voice.

The results of this model call into question the anticipated parallelism between the F1 nucleus and F2 nucleus models: while speakers consistently trend toward backing /aw/ nuclei over apparent time, they also vary in how much they concomitantly lowered /aw/ nuclei (as would be expected, physiologically, for backing in the lower part of the vowel space). In particular, women and white collar speakers appear to lag behind men and blue collar peers in nucleus lowering, although all groups back nuclei comparably. If the relationship between nucleus height and backness were purely structural, physiologically speaking, the observed level of variability in nucleus height conditioned by social factors would not be expected. What this conditioned

variability may point to is a separate sort of physiological force acting on nucleus height.

4.3. F1 Glide Model

The log-likelihood comparisons for the F1 glide model, shown in Table 4, reveal a significant main effect of BIRTH YEAR: younger speakers have lower F1 glide values (i.e., more raising) than older speakers. Figure 4 shows the linear relationship between BIRTH YEAR and F1 glide values, in which the youngest speakers have the lowest F1 glide values. That is, younger speakers have higher glide targets for /aw/ than older speakers do.

Table 4. Log-likelihood Comparisons for the Linear Mixed-Effects Models With F1 Glide as Dependent Variable

Coefficient	χ^2 Value	Degrees of freedom	p Value
BIRTH YEAR $\times$ GENDER $\times$ CLASS	0.859	1	n.s.
BIRTH YEAR $\times$ GENDER	1.191	1	n.s.
BIRTH YEAR $\times$ CLASS	2.887	1	n.s.
GENDER $\times$ CLASS	0.128	1	n.s.
BIRTH YEAR	52.961	1	<.001
GENDER	11.784	1	.001
CLASS	1.545	1	n.s.
FOLLOWING VOICE	28.84	1	<.001
DURATION	6.50	1	0.011

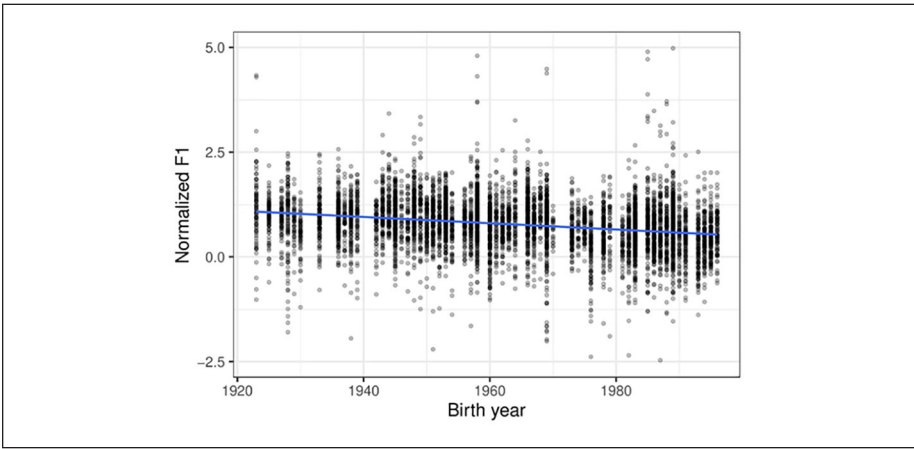


Figure 4. Linear Effect of BIRTH YEAR on F1 Glide Values

The results shown in Table 4 also demonstrate a significant main effect of GENDER: women have lower F1 glide values (i.e., more raised glides) than men. Since the effect of BIRTH YEAR in the model suggests an overall trend of glide raising for all speakers over apparent time, this result indicates that women lead same-aged men in glide raising. However, unlike the F1 nucleus model, there is no significant main effect of CLASS.

Both the FOLLOWING VOICE and DURATION covariates in this model are significant. Longer vowel durations are associated with more raised glides, and pre-voiced tokens (e.g., *clout*) have more raised glides than pre-voiceless (e.g., *cloud*) tokens on average. The duration result is likely due to the increased time necessary to reach a target that is more raised (i.e., further from the nucleus target).

The results of this model support my prediction that Raleigh speakers are participating in a retreat from the lowered glides that are typical of Southern speakers (Thomas 2003). In this way, changes in glide height parallel Raleigh speakers' retreat from other canonically Southern vocalic features, such as the front vowel components of the SVS (Dodsworth & Kohn 2012). This glide lowering trend is led by women, although there is no class stratification in the uptake of lowering.

5. Discussion

5.1. Formant Dynamics and Relationships Among Ongoing Changes in /aw/

A summary of the statistical results is shown in Table 5. Although the three statistical models have different dependent variables, they are necessarily related to one another because their dependent variables are not independent from one another due to being measurements of individual diphthong productions. Therefore, the relationships among the different statistical models form the basis for my interpretation of the overall results.

Table 5. Practical Summaries of Significant Effects and their Directions

Acoustic parameter	Significant main effects	Directionality and leader of change over apparent time
F1 nucleus	BIRTH YEAR GENDER CLASS	Lowering, led by men and blue collar speakers
F2 nucleus	BIRTH YEAR	Backing
F1 glide	BIRTH YEAR GENDER	Raising, led by women

These results point to all speakers both lowering and backing /aw/ nuclei over apparent time, with women and white collar speakers raising slightly relative to phonetically natural lowering. An additional piece of evidence supporting this interpretation is that women also lead glide raising, which may be pulling nucleus height with it and creating the gender effect observed in the F1 nucleus model. That is, the apparent lag in the uptake of nucleus lowering by women can be attributed to another process that appears to operate semi-independently of the backing dimension of /aw/ retraction.

Under this analysis, there are two (opposing) forces acting upon nucleus height simultaneously: a lowering force via nucleus backing and a raising force via glide raising. In the course of /aw/ retraction, the nucleus tends to both lower and back, as has been noted in other related studies (Strelluf 2018), because the lower part of the vowel space is smaller. This physiological restriction is the consequence of human mouths having less space for tongue movement on the front-back dimension in the bottom part of the mouth than in the top part of the mouth. Therefore, nucleus backing tends to entail nucleus lowering.

At the same time, glide raising exerts coarticulatory pressure on the nucleus to also raise. Since /aw/ is a diphthong with two acoustic targets, one for the nucleus and another for the glide, changes in the glide target can affect the nucleus target (and vice versa) because the two targets must be reached in quick succession to form a single vowel. Given this time constraint, these two targets are frequently coarticulated and thus influence one another. Therefore, a raised glide tends to also raise the nucleus target to reduce the acoustic distance between the two targets overall.

Another important element of this interpretation is duration and how it interacts with various acoustic realizations. Longer tokens have fronter and lower nuclei, along with more raised glides, than shorter tokens. These results can be interpreted to mean that spectral rate of change is relatively stable, since duration increases as nuclei lower and glides raise. If duration were insensitive to formant dimensions, that would indicate that the structural distance between the nucleus and glide targets is a stable element in diphthong production. In future research, close examination of the spectral rate of change, involving more measurements per vowel, would be informative.

The shape of the /aw/ diphthong trajectory, especially in terms of its height dimension, is changing over apparent time, as shown in Figure 5. This figure treats birth year categorically for the sake of visual clarity, following Dodsworth and Kohn's (2012) generational divisions for the same corpus. The three generations are split by the amount and nature of dialect contact that participants grew up with in the Raleigh area. Generation 1 speakers (born 1923-1954) completed at least elementary school prior to Research Triangle Park's takeoff, generation 2 speakers (born 1955-1978) were the first group to grow up among children whose parents had moved from out of state, and generation 3 speakers (born 1979-1996) grew up in an even later stage of this dialect contact situation.

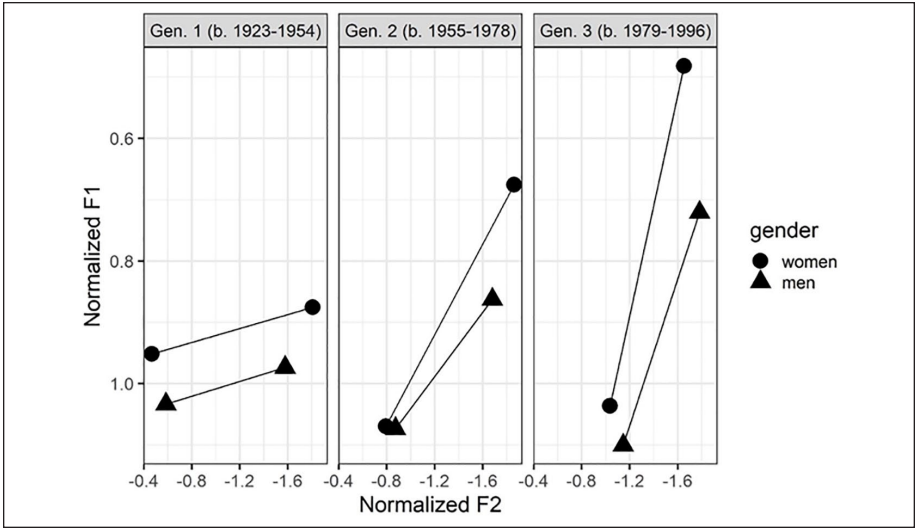


Figure 5. Mean Formant Trajectories by GENDER and GENERATION

For the oldest speakers, most of the distance between the nucleus and glide targets is accounted for by changes in F2. However, by generation 2, the distance between the nucleus and the glide targets appears to be more evenly distributed between F1 and F2. Generation 2 speakers show the middle stage of the shift toward larger F1 displacement. Finally, by generation 3, the majority of the distance between the nucleus and the glide targets is attributable to changes in F1. Nucleus retraction is accompanied by wholesale changes to the shape of the diphthong's trajectory, reinforcing the notion that speakers are not targeting a specific sort of consistent structural relationship between nucleus and glide targets when producing /aw/. This trend is also evident in the average acoustic distance between nuclei and glides by generation and gender, summarized in Table 6 which shows that glide lengths are increasing over apparent time with women leading the change.

Table 6. Means and Standard Deviations for Glide Lengths (Lobanov-Normalized Units) Between 25 percent and 75 percent Acoustic Measurements by GENERATION and GENDER

Generation	Gender	
	Women	Men
1 (1923-1954)	0.853 (0.240)	0.824 (0.310)
2 (1955-1978)	1.017 (0.397)	0.869 (0.321)
3 (1979-1996)	1.361 (0.292)	1.168 (0.289)

5.2. Placing /aw/ in the Larger Context of Vocalic Changes in Raleigh

The effects of social factors such as GENDER and CLASS on changes in /aw/ are variable. My analyses show that women are leading glide raising, along with lagging in nucleus

lowering. In section 5.1, I proposed that these two changes are likely physiologically related to one another, such that raised glides are bringing otherwise lowered nuclei upwards just a bit. This interpretation puts women at the forefront of glide raising (and its resonant effects on nucleus height), which constitutes movement away from the lowered glide that is typical of more Southern /aw/ productions (Thomas 2008). Labov, Rosenfelder, and Fruehwald (2013) report that younger Philadelphians retract fronted /aw/ nuclei to sound less Southern, which is a different acoustic parameter of /aw/. However, it seems possible that reversal of /aw/ glide lowering may also serve this aim given that both features are characteristic of Southern English (Thomas 2008) and given that this reversal began among speakers born in the 1950s in both Philadelphia and Raleigh.

This interpretation of the data aligns with Labov's (1990) claim that women tend to lead in the uptake of incoming variants, as raised glide variants are likely introduced into the community by Northerners moving in. However, it is worth noting that these variants also exist in the community, albeit relatively rarely, among speakers of older generations. Figure 6 shows mean normalized glide height for all speakers (black = women; white = men). Raised glide variants (points closer to the top of the plot) occur in a wide range of birth years, especially among white collar women shown with black triangles. In contrast, lowered glide variants (points closer to the bottom of the plot) also occur in a wide range of birth years, and disproportionately among men, shown in gray. The existence of the incoming variant, the raised glide, in the community prior to its widespread uptake mirrors Dodsworth and Kohn's (2012) observations about the reversal of SVS-related front vowel shifts among speakers in the Raleigh corpus. They reported that the incoming variants (i.e., less Southern-shifted front vowels) were produced by a few older speakers, pointing to substantial variability in the pool of vocalic variation even preceding extensive in-migration.

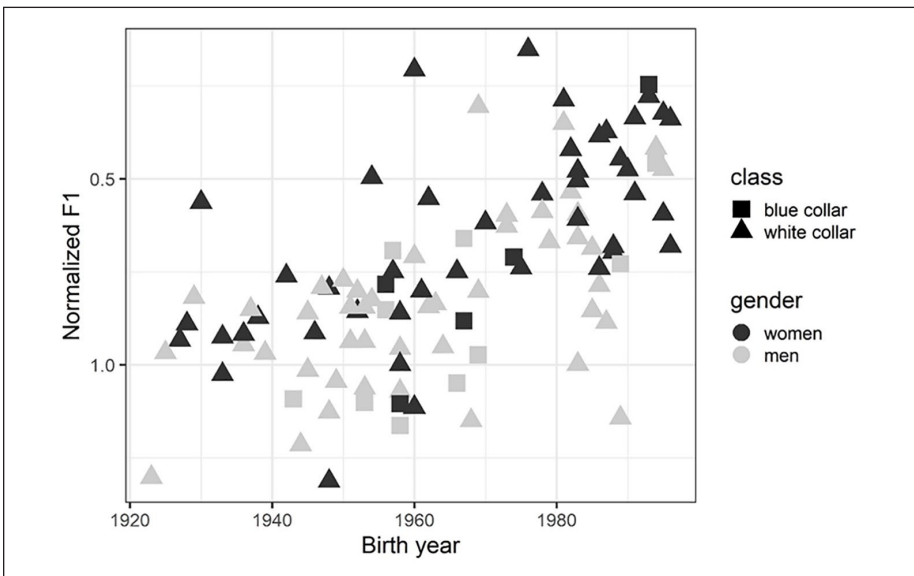


Figure 6. Mean F1 Glide Values for All Speakers

However, it is also worth noting that /aw/ is a low salience feature in white Southern English relative to, for example, /aɪ/ monophthongization (Thomas 2003). Therefore, it is possible that the changes in /aw/ in Raleigh reflect adoption of an “exonorm” (DuBois & Horvath 1999), that is, a norm that originates from outside a community, that gets stronger with each passing generation. This process would be consistent with other vocalic changes that have been documented among white Raleigh residents, as well, in terms of orientation away from white Southern English.

As for the progression of the reversal of /aw/ fronting in this community, it appears that Raleigh speakers’ /aw/ productions pattern with other speakers in the Southeastern super-region as reported by Labov, Rosenfelder, and Fruehwald (2013) for Philadelphia and Strelluf (2018) for Kansas City. In addition to providing evidence of this reversal in a Southern urban area within this larger super-region, this data also suggests that /aw/ fronting may not remain a defining feature of the super-region for much longer. More research is needed to understand the full extent of the reversal of /aw/ fronting in the Southeastern super-region, especially in less urban areas.

The lack of social stratification in this reversal also prompts questions about its origins, and crucially whether they differ from the origins of /aw/ glide raising in the same speakers. While I have proposed that the social stratification of the uptake of /aw/ glide raising points to raised glides as an incoming variant in the greater Raleigh community (Labov 1990) or perhaps orientation toward an exonorm (DuBois & Horvath 1999), the uptake of the reversal of /aw/ fronting was not significantly impacted by any social factor other than birth year. It is possible that the nucleus of /aw/ has been reanalyzed as the low front vowel /æ/, as was proposed for white Kansas Citians by Strelluf (2018). He theorizes that such a reanalysis, in combination with /æ/ retraction, could prompt /aw/ retraction by phonetic analogy. Dodsworth and Kohn (2012) did report /æ/ retraction among younger white speakers in the Raleigh corpus, so an analogy likening /æ/ retraction to /aw/ retraction would be plausible if speakers were to phonologically reanalyze the nucleus of /aw/ as /æ/. Although additional investigation of this possibility is certainly warranted, concurrent /æ/ and /aw/ retraction in both Kansas City and Raleigh is a promising dialectological parallel that may point to a broader phonological trend in the Southeastern super-region. Such a trend would indicate that the structural motivation for /aw/ retraction is crucially shared by speakers far away from one another, suggesting that /aw/ retraction can be independently innovated via phonological reanalysis in communities with minimal contact with one another.

My analysis provides additional supporting evidence that /aw/ retraction in the Southeastern super-region could be structurally motivated in speakers’ phonologies, much like what has been proposed for other chain shifts by Labov (1994). Perhaps /æ/ retraction, in combination with fronted /aw/ nuclei in the same speaker, creates an ideal condition for that speaker to phonologically reassign the nucleus of /aw/. Dinkin (2013) made a similar conceptual argument in favor of pre-lateral /æ/ (e.g., *pal*) being phonologically reassigned to /aw/ among Philadelphians. In other words, a mechanism of phonological reassignment in response to appropriate structural conditions in the

vowel system could be sensitive to dialect-driven phonetic similarities among phonological categories.

6. Conclusion

The differences in the uptake of nucleus backing and glide raising do suggest that these two changes are at minimum somewhat structurally independent from one another, and the trend in the nucleus height data is consistent with an acoustic parameter that is being pulled in opposite directions (upwards by glide raising, but downwards by nucleus retraction). I argue that glide raising is more likely to be motivated by social factors than nucleus retraction, given the way it is stratified in my data set and how its timing aligns with other changes away from the SVS among Raleigh speakers and speakers elsewhere in the Southeastern super-region. Since the backing aspect of nucleus retraction is broadly taken up in the community across social divisions, this change in progress seems structurally motivated. That these two opposing forces (nucleus lowering due to nucleus backing, and nucleus raising due to glide raising) appear to be active in the community at the same time offers a dynamic look at how phonological and social forces interact with one another to create complex vocalic patterns.

Whether speakers in Raleigh are moving away from the SVS or toward a supra-regional pattern is challenging to differentiate, especially absent explicit metalinguistic commentary on /aw/ from people in the corpus, but the vocalic patterns in this data set offer relevant clues to what may be happening. Glide raising likely constitutes a movement away from the SVS, since its uptake (and corresponding effect on nucleus height) is socially stratified such that women and white collar speakers are the innovators. The stratification of this pattern closely mirrors findings about gender-based and class-based differences in the uptake of sound changes from above, both in Raleigh (Dodsworth & Kohn 2012) and elsewhere in western contexts (Labov 1990, 1994).

On the other hand, I do not find evidence that the uptake of /aw/ retraction is socially stratified in a similar way, suggesting it may have a different relationship to local versus supra-regional vocalic norms. While it is tempting to propose that /aw/ retraction is a novel supra-regional vocalic norm for the Southeastern super-region, at least in urban areas, it is less clear whether it makes sense to think about /aw/ retraction as orientation toward or away from any sort of norm. If the /aw/ vowel class is indeed subject to phonological reassignment due to its phonetic characteristics, speakers themselves aren't really doing any orienting toward or away from particular norms; rather, something cognitive has shifted resulting in a structural change to their phonological systems. In this sense, the result of phonological reassignment may outwardly appear to constitute speaker orientation toward some new supra-regional norm of /aw/ retraction in the Southeastern super-region (especially when /aw/ retraction has been reported in other cities, like Philadelphia and Kansas City), but in reality it may simply occur due to appropriate structural conditions present within speakers' phonologies that are shared in multiple parts of the super-region.

The confluence of these two changes in Raleigh speakers' /aw/ productions has produced patterns that exemplify how changes with variable motivations can interact with one another to complex ends. While it remains to be seen whether these patterns are unique to Raleigh or a broader trend in urban areas in the American South, they provide reasons to carefully reconsider the hallmark characteristics of the Southeastern super-region, as well as whether shared changes in urban areas throughout constitute shared structural motivations for sound change. Additional findings pertaining to the concomitance of /æ/ retraction and /aw/ retraction have the potential to further bolster the argument that supra-regional level sound changes are structurally connected to one another.

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Software

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